

# A Level Mechanics

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## Chapter 4: Variable Acceleration

### Why this chapter matters

In Chapter 2, acceleration was always constant, so the suvat equations could describe the whole motion in one line. Real motion is rarely that tidy — acceleration itself often changes with time — and once it does, suvat no longer applies at all. This chapter replaces suvat with calculus: differentiation and integration link displacement, velocity and acceleration directly, and become the standard tools whenever acceleration (or velocity) is given as a function of time rather than as a constant.

This is a genuine step up in *how* you think about kinematics, not just a new formula to memorise. Under suvat, velocity and acceleration were single numbers that stayed fixed for the whole motion. Here,  $s$ ,  $v$  and  $a$  are all *functions of time* — they can be evaluated at any instant, and a question is often asking you to find a specific instant (when is the particle at rest? when is velocity greatest?) rather than to describe the whole journey at once. The calculus itself — differentiating and integrating powers of  $t$  — should already feel familiar from Pure Mathematics; what's new is applying it to a physical, changing quantity and interpreting the result in context. Expect this chapter to take more practice than Chapter 2, precisely because each question now blends two skills: the calculus, and the physical interpretation of what you've just calculated.

### 4.1 Differentiating to find velocity and acceleration

#### Concept

When displacement is given as a function of time,  $s(t)$ , velocity and acceleration are found by differentiating:

Because each derivative is itself a function of  $t$ , velocity and acceleration can be evaluated at any instant — not just found as single constant values the way they were under suvat.

- The **sign of  $v(t)$**  tells you the *direction* of motion (positive or negative, relative to whichever direction you chose as positive).
- The **sign of  $a(t)$**  tells you whether the particle is *speeding up or slowing down* in that direction.

Keep those two readings separate: a positive velocity does not by itself tell you anything about acceleration, and vice versa. You need both together to describe the motion fully, as in the worked example below.

#### Key vocabulary and notation

| Term   | Meaning                                |
|--------|----------------------------------------|
| $s(t)$ | Displacement as a function of time $t$ |

|                                                                   |                                                                                                                      |
|-------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|
| <b>v(t)</b>                                                       | Velocity function, $v = ds/dt$                                                                                       |
| <b>a(t)</b>                                                       | Acceleration function, $a = dv/dt = d^2s/dt^2$                                                                       |
| <b>Derivative</b>                                                 | The rate of change of one quantity with respect to another — here, always with respect to time                       |
| <b>Rate of change</b>                                             | How quickly a quantity changes per unit time; found by differentiating                                               |
| <b><math>\dot{s}</math>, <math>\ddot{s}</math> (dot notation)</b> | Alternative notation sometimes seen: $\dot{s}$ means $ds/dt$ (velocity), $\ddot{s}$ means $d^2s/dt^2$ (acceleration) |

### Key formulas

#### KEY FORMULA

$$v = ds/dt$$

$$a = dv/dt = d^2s/dt^2$$

Both are found using the standard power rule from Pure Mathematics: if  $s = kt^n$ , then  $ds/dt = nkt^{n-1}$ . There is no new differentiation *technique* here — what's new is applying differentiation to kinematics in the first place, and remembering to interpret the number you get.

### Procedure

1. Write down  $s(t)$  exactly as given, term by term.
2. Differentiate once, term by term, using the power rule, to obtain  $v(t)$ .
3. Differentiate again (or differentiate  $v(t)$  once more) to obtain  $a(t)$ .
4. Substitute the required value of  $t$  into  $v(t)$  and/or  $a(t)$ .
5. Interpret the sign of each result in context: direction of motion, and whether the particle is speeding up or slowing down.

### Worked example

#### EXAMPLE 4.1

**Question.** A particle moves in a straight line so that its displacement from a fixed point O,  $s$  metres, at time  $t$  seconds, is given by  $s = t^3 - 6t^2 + 9t$ , for  $t \geq 0$ . Find (a) the velocity and acceleration of the particle when  $t = 2$ , and (b) describe the motion of the particle at this instant.

**Step 1 — differentiate  $s$  to find  $v$ .** Differentiate each term of  $s = t^3 - 6t^2 + 9t$  using the power rule (multiply by the power, then reduce the power by 1):

$$v = ds/dt = 3t^2 - 12t + 9$$

**Step 2 — differentiate  $v$  to find  $a$ .** Either differentiate  $v(t)$  again, or differentiate  $s(t)$  twice — both give the same result:

$$a = dv/dt = 6t - 12$$

**Step 3 — substitute  $t = 2$  into  $v(t)$  and  $a(t)$ .** Work through the arithmetic one operation at a time

so no sign is dropped:

$$v(2) = 3(2)^2 - 12(2) + 9 = 3(4) - 24 + 9 = 12 - 24 + 9 = -3 \text{ m s}^{-1}$$

$$a(2) = 6(2) - 12 = 12 - 12 = 0 \text{ m s}^{-2}$$

**Step 4 — interpret both signs.** This is the step most often skipped, but it usually carries its own mark:

**Answer:** (a) At  $t = 2$ ,  $v = -3 \text{ m s}^{-1}$  and  $a = 0 \text{ m s}^{-2}$ . (b) The particle is moving in the negative direction at this instant, and its velocity is momentarily stationary (neither increasing nor decreasing).

### Why this matters beyond the algebra

#### WHY THIS MATTERS BEYOND THE ALGEBRA

Differentiating is the easy part once the power rule is secure — the skill this section is really testing is treating  $v(t)$  and  $a(t)$  as *functions* you can evaluate and interpret, rather than as single fixed numbers. Two habits are worth building from your very first example:

### Common exam mistakes

- Finding  $v(t)$  or  $a(t)$  correctly but stopping without substituting the required value of  $t$ .
- Reporting a velocity or acceleration as a bare number without interpreting its sign (direction, speeding up/slowing down).
- Confusing "acceleration is zero" with "the particle is at rest" — these are different conditions ( $a = 0$  vs  $v = 0$ ); this distinction is central to Section 4.3.
- Differentiation slips — e.g. dropping a coefficient or a sign when a term is negative — from being new to applying the power rule inside a kinematics context.

## 4.2 Integrating to find velocity and displacement

### Concept

Integration reverses differentiation, so it's the tool for working backwards: from acceleration to velocity, and from velocity to displacement.

Because integration is the reverse process, every **indefinite** integral introduces an unknown constant of integration, "+C", which must be found using a given initial condition — a known value of  $v$  or  $s$ , usually at  $t = 0$ . A **definite** integral, evaluated between two specific times, gives the exact *change* in velocity or displacement over that interval, and needs no constant at all, because the constants cancel when you subtract.

### Key vocabulary and notation

| Term | Meaning |
|------|---------|
|------|---------|

| <b>Indefinite integral</b>                          | $\int a \, dt$ or $\int v \, dt$ — produces a function plus an unknown constant, "+C"                                    |
|-----------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|
| <b>Constant of integration</b>                      | The unknown constant introduced by indefinite integration; found from an initial condition                               |
| <b>Initial condition</b>                            | A given value of $v$ or $s$ at a specific time (often $t = 0$ ), used to find the constant                               |
| <b>Definite integral</b>                            | An integral evaluated between two limits, e.g. from $t_1$ to $t_2$ — gives a specific numerical change, with no constant |
| <b><math>\Delta v</math>, <math>\Delta s</math></b> | Change in velocity / change in displacement over a time interval                                                         |

### Key formulas

#### KEY FORMULA

$$v(t) = \int a(t) \, dt$$

$$s(t) = \int v(t) \, dt$$

$$\Delta v = \int a(t) \, dt \text{ from } t_1 \text{ to } t_2, \quad \Delta s = \int v(t) \, dt \text{ from } t_1 \text{ to } t_2$$

### Procedure

1. Integrate the given function term by term, using the reverse power rule, and add a constant "+C".
2. Substitute the given initial condition (a known value of  $v$  or  $s$  at a known  $t$ ) to find  $C$ .
3. Rewrite the function with  $C$  replaced by its value.
4. For a change over an interval, either substitute the two limits into the now-complete function and subtract, or evaluate the definite integral directly — both give the same answer, and no constant is needed for a definite integral.

### Worked example

#### EXAMPLE 4.2

**Question.** A particle moves in a straight line so that its acceleration at time  $t$  seconds is  $a = (4t - 3) \, \text{m s}^{-2}$ . When  $t = 0$ , the particle has velocity  $v = 2 \, \text{m s}^{-1}$  and displacement  $s = 5 \, \text{m}$  from a fixed point  $O$ . Find (a)  $v(t)$  and  $s(t)$ , and (b) the displacement of the particle between  $t = 1$  and  $t = 3$ .

**Step 1 — integrate  $a(t)$  to find  $v(t)$ .** Integrate each term of  $a = 4t - 3$  using the reverse power rule (raise the power by 1, then divide by the new power), and remember the constant:

$$v = \int (4t - 3) \, dt = 4t^2/2 - 3t + C = 2t^2 - 3t + C$$

**Step 2 — use  $v = 2$  when  $t = 0$  to find  $C$ .** Substitute  $t = 0$ ,  $v = 2$  into the equation from Step 1:

$$2 = 2(0)^2 - 3(0) + C \implies 2 = C \implies C = 2$$

So  $v(t) = 2t^2 - 3t + 2$ .

**Step 3 — integrate  $v(t)$  to find  $s(t)$ .** This is a *second, separate* integration, so it needs its *own*,

separate constant — call it  $D$  so it isn't confused with  $C$ :

$$s = \int (2t^2 - 3t + 2) dt = 2t^3/3 - 3t^2/2 + 2t + D$$

**Step 4 — use  $s = 5$  when  $t = 0$  to find  $D$ .** Substitute  $t = 0$ ,  $s = 5$ :

$$5 = 2(0)^3/3 - 3(0)^2/2 + 2(0) + D \implies 5 = D \implies D = 5$$

So  $s(t) = (2/3)t^3 - 1.5t^2 + 2t + 5$ .

**Step 5 (part b) — find the displacement between  $t = 1$  and  $t = 3$ .** Either evaluate  $s(t)$  at both limits and subtract, or integrate  $v(t)$  between the limits directly — here we use the first method, since  $s(t)$  is already known:

$$s(3) = 2(3)^3/3 - 3(3)^2/2 + 2(3) + 5 = 18 - 13.5 + 6 + 5 = 15.5$$

$$s(1) = 2(1)^3/3 - 3(1)^2/2 + 2(1) + 5 = 2/3 - 1.5 + 2 + 5 = 37/6$$

$$\Delta s = s(3) - s(1) = 15.5 - 37/6 = 28/3 \approx 9.33 \text{ m}$$

**Answer:** (a)  $v(t) = 2t^2 - 3t + 2$ , and  $s(t) = (2/3)t^3 - 1.5t^2 + 2t + 5$ . (b) The displacement between  $t = 1$  and  $t = 3$  is  $28/3 \text{ m} \approx 9.33 \text{ m}$ .

### Why two separate constants matter

#### WHY TWO SEPARATE CONSTANTS MATTER

Integrating a kinematics function is a genuinely different skill from differentiating one, not just the same idea run in reverse — because integration always leaves an unknown behind that you have to go and find. Two things are worth being deliberate about:

- Make sure the reverse power rule ( $\int t^n dt = t^{n+1}/(n+1) + C$ ) is completely automatic as pure algebra *before* applying it here, so you can focus on the physics, not the mechanics of integrating.
- Every *new* indefinite integral needs a *new* constant, found from its *own* condition. Example 4.2 needs two separate constants ( $C$  for velocity,  $D$  for displacement), found from two separate conditions — mixing them up (for instance, using the displacement condition to find  $C$ ) is one of the most common slips at this stage.

It's also worth flagging early: a definite integral of  $v(t)$  over an interval gives *displacement*, and displacement only equals *distance travelled* if the particle doesn't change direction within that interval. Section 4.3 shows what to do when it does.

### Common exam mistakes

- Forgetting the "+C" entirely, or forgetting to find its value from the initial condition.
- Using the wrong initial condition for the wrong constant (e.g. substituting the value of  $s$  into the equation for  $v$ ).
- Sign errors when integrating a term that is already negative.
- Assuming a definite integral of  $v(t)$  over an interval always gives total distance travelled — it gives displacement, which understates distance whenever the particle reverses direction within that interval.

### 4.3 Stationary values and distance travelled

#### Concept

Calculus also answers questions suvat can't: where and when is velocity at a maximum or minimum, when is the particle momentarily at rest, and how far does it actually travel — as opposed to its net displacement?

- A **stationary value of velocity** occurs where  $dv/dt = a(t) = 0$  — a turning point of the  $v(t)$  graph — and is classified as a maximum or minimum using the second derivative ( $d^2v/dt^2 = da/dt$ ) or a sign test either side of that value.
- A particle is **momentarily at rest**, and may be changing direction, wherever  $v(t) = 0$ . (This is also a stationary point of the displacement graph, since  $ds/dt = 0$  there.)
- **Total distance travelled** differs from displacement whenever the particle changes direction. Distance is found by splitting the time interval at every point where  $v(t)$  changes sign, finding the size (magnitude) of the displacement over each piece, and adding those magnitudes together — equivalent to integrating  $|v(t)|$ .

The key idea to hold onto: " $v = 0$ " and " $dv/dt = 0$ " are two different conditions asking two different questions (is the particle at rest? vs is velocity at a turning point?), and mixing them up is the single most common error in this section.

#### Key vocabulary and notation

| Term                                        | Meaning                                                                                                               |
|---------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|
| <b>Stationary point (of <math>v</math>)</b> | A value of $t$ where $dv/dt = a = 0$ ; $v(t)$ has a turning point there                                               |
| <b>Particle at rest</b>                     | $v(t) = 0$ — the particle is momentarily stationary, and may reverse direction                                        |
| <b>Change of direction</b>                  | Occurs where $v(t)$ passes through zero and changes sign (not merely equals zero)                                     |
| <b>Maximum / minimum velocity</b>           | A stationary value of $v(t)$ , classified using the second derivative or a sign test                                  |
| <b>Speed</b>                                | $ v(t) $ , the magnitude of velocity — always non-negative                                                            |
| <b>Total distance travelled</b>             | The sum of the magnitudes of displacement over each sub-interval where $v(t)$ keeps one sign; equals $\int  v(t)  dt$ |

#### Key formulas

##### KEY FORMULA

$$dv/dt = 0 \implies \text{stationary point of } v(t) \text{ (max/min velocity)}$$

$$v(t) = 0 \implies \text{particle momentarily at rest}$$

$$\text{Total distance} = \int |v(t)| dt \text{ from } t_1 \text{ to } t_2 = \sum |\Delta s| \text{ over intervals where } v \text{ keeps one sign}$$

### Procedure

1. To find when the particle is at rest, solve  $v(t) = 0$ .
2. To find maximum or minimum velocity, solve  $a(t) = dv/dt = 0$ , then classify each solution using  $d^2v/dt^2$  (positive result = minimum, negative result = maximum) or by checking the sign of  $a(t)$  either side.
3. To find total distance travelled over  $[t_1, t_2]$ , find every value of  $t$  in that interval where  $v(t) = 0$ . These values split the interval into pieces on which  $v$  does not change sign.
4. Evaluate  $s(t)$  (found by integration, as in Section 4.2) at every endpoint of every piece, take the magnitude of the change across each piece, and add all the magnitudes together.

### Worked example

#### EXAMPLE 4.3

**Question.** A particle moves in a straight line so that its velocity at time  $t$  seconds is  $v = t^2 - 4t + 3$  ( $\text{m s}^{-1}$ ), for  $0 \leq t \leq 4$ . When  $t = 0$ , the particle's displacement from O is 0. Find (a) the times when the particle is momentarily at rest, (b) the minimum velocity in this interval, showing that it is a minimum, and (c) the total distance travelled between  $t = 0$  and  $t = 4$ .

**Step 1 (part a) — set  $v = 0$  and solve.** Factorise the quadratic:

$$t^2 - 4t + 3 = 0 \implies (t - 1)(t - 3) = 0 \implies t = 1 \text{ or } t = 3$$

**Step 2 (part b) — differentiate  $v$  to find  $a$ , and set  $a = 0$ .** Remember: this is a *different* equation from Step 1 — here we're looking for a turning point of velocity, not for when velocity is zero:

$$a = dv/dt = 2t - 4 = 0 \implies t = 2$$

**Step 3 — find  $v(2)$ , and classify it using the second derivative.**

$$v(2) = (2)^2 - 4(2) + 3 = 4 - 8 + 3 = -1 \text{ m s}^{-1}$$

$$d^2v/dt^2 = 2 > 0 \implies \text{minimum}$$

**Step 4 (part c) — integrate  $v$  to find  $s(t)$ , then evaluate  $s$  at each split point.** Use  $s(0) = 0$  to find the constant, then evaluate  $s$  at  $t = 0, 1, 3, 4$  — the rest points from part (a) are exactly where  $v$  changes sign, so they split  $[0, 4]$  into pieces of constant sign:

$$s(t) = \int (t^2 - 4t + 3) dt = t^3/3 - 2t^2 + 3t + E, \quad s(0) = 0 \implies E = 0$$

$$s(t) = t^3/3 - 2t^2 + 3t$$

$$s(0) = 0, \quad s(1) = 1/3 - 2 + 3 = 4/3, \quad s(3) = 9 - 18 + 9 = 0, \quad s(4) = 64/3 - 32 + 12 = 4/3$$

**Step 5 — sum the magnitude of the displacement across each piece.** Between the rest points,  $s$  changes monotonically, so the distance covered in each piece is just the size of the change in  $s$ :

$$\text{distance} = |s(1) - s(0)| + |s(3) - s(1)| + |s(4) - s(3)| = |4/3| + |-4/3| + |4/3| = 4/3 + 4/3 + 4/3 = 4 \text{ m}$$

**Answer:** (a) The particle is at rest at  $t = 1$  s and  $t = 3$  s. (b) The minimum velocity is  $-1 \text{ m s}^{-1}$ , at  $t = 2$  s, confirmed by  $d^2v/dt^2 = 2 > 0$ . (c) Total distance travelled = 4 m. Note this is *three times* the net displacement over the whole interval,  $s(4) - s(0) = 4/3$  m, because the particle changes direction twice — a useful sanity check that distance and displacement really are different

quantities here.

### Why this section is tricky

#### WHY THIS SECTION IS TRICKY

This is often the hardest part of the chapter, even once Sections 4.1 and 4.2 feel comfortable, because it asks for a *strategy* built from several earlier ideas rather than a single formula to apply. The biggest conceptual trap is mixing up two similar-looking conditions: " $v = 0$ " finds when the particle is at rest, while " $dv/dt = 0$ " finds a turning point of velocity itself — they are rarely satisfied by the same value of  $t$ , and a question will often ask for both in the same context (as Example 4.3 does).

A strong habit to build here: sketch a rough  $v$ - $t$  graph before attempting any distance question. Even a rough sketch makes the sign changes — and therefore the points where the interval needs to be split — visually obvious, and it heads off the single most common error in this section (see below).

### Common exam mistakes

- Setting  $dv/dt = 0$  to find when the particle is at rest, instead of setting  $v = 0$  ( $dv/dt = 0$  finds stationary values of velocity, not rest).
- Computing total distance by evaluating a single definite integral of  $v(t)$  over the whole interval, without checking for sign changes first — this silently computes displacement, not distance, whenever the particle reverses direction.
- Finding a stationary value of  $v(t)$  but failing to justify whether it's a maximum or minimum (no second derivative or sign test shown), losing a method or accuracy mark.
- Sign errors when combining  $|\Delta s|$  across pieces, especially forgetting to take the magnitude and ending up with a "distance" smaller than the net displacement — a result that should immediately look suspicious.

### Chapter summary

| Idea                                                                    | Key point                                                                                        |
|-------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|
| $v = ds/dt$ , $a = dv/dt = d^2s/dt^2$                                   | Differentiate displacement once for velocity, twice (or differentiate velocity) for acceleration |
| $v(t) = \int a \, dt$ , $s(t) = \int v \, dt$                           | Integrate acceleration/velocity; always add "+C" and use an initial condition to find it         |
| <b>Definite integral of <math>v</math> over <math>[t_1, t_2]</math></b> | Gives the signed change in displacement over that interval — not necessarily distance            |
| <b>Particle at rest</b>                                                 | $v(t) = 0$                                                                                       |
| <b>Max/min velocity</b>                                                 | $a(t) = dv/dt = 0$ ; classify with $d^2v/dt^2$ or a sign test                                    |

|                       |                                                                                                 |
|-----------------------|-------------------------------------------------------------------------------------------------|
| <b>Total distance</b> | Split the interval at every point where $v$ changes sign, then sum $ \Delta s $ over each piece |
|-----------------------|-------------------------------------------------------------------------------------------------|

**Common exam mistakes (chapter-level)**

- Confusing distance with displacement, and confusing "at rest" ( $v = 0$ ) with "zero acceleration" ( $a = 0$ ).
- Forgetting the constant of integration, or applying the wrong initial condition to find it.
- Assuming  $\int v \, dt$  over an interval automatically gives total distance travelled, without checking for a change of direction.
- Sign errors whenever a particle changes direction, especially in distance-travelled calculations.
- Treating this chapter as a minor add-on rather than core content — variable acceleration is examined every year and builds directly on the calculus you already know from Pure Mathematics.

**You should now be able to**

- Differentiate a displacement function  $s(t)$  to find velocity, and differentiate again (or differentiate  $v(t)$ ) to find acceleration, interpreting the sign of each in context.
- Integrate a given acceleration or velocity function, using initial conditions to evaluate the constant of integration.
- Use definite integration to find the change in velocity or displacement over a given time interval.
- Find when a particle is at rest ( $v = 0$ ) and when its velocity is a maximum or minimum ( $dv/dt = 0$ ), and justify which using the second derivative or a sign test.
- Distinguish total distance travelled from displacement, and calculate total distance using  $|v|$  when the particle changes direction within the interval.